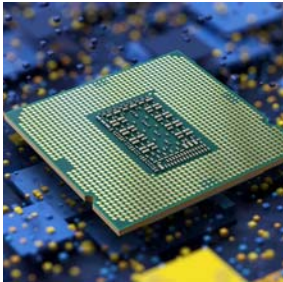


A 1.1.1 The CPU



A 1.1.1 Describe the functions and interactions of the main CPU components.

The central processing unit (CPU) carries out every instruction a computer runs, from a single addition to the millions of operations behind a video call. It contains a small set of components, each with one clear job, and the way those components pass data and signals between them is what makes a program run. The syllabus names every component you need to know: two units, five registers, three buses and three types of processor. This page describes the function of each component, explains how they interact, and ends with the labelled diagram the syllabus requires you to know.

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Key Questions

- What do the ALU and the control unit each do?
- What does each of the five registers (IR, PC, MAR, MDR, AC) hold, and why does the CPU need them?
- What travels along the address, data and control buses, and in which direction?
- How do the components interact when the CPU carries out an instruction?
- How does a multi-core processor differ from a single-core processor, and what is a co-processor for?
- Can you draw and label a diagram showing how these components connect?

IB Command Term note

DESCRIBE: Give a detailed account. For each component: state what it is, what it holds or does, and how it connects to the other components.

WHAT TO INCLUDE: Every item the syllabus names: the ALU and CU; the five registers (IR, PC, MAR, MDR, AC); the three buses (address, data, control); single-core processors, multi-core processors and co-processors; and the diagram showing how they relate.

The two units: the ALU and the control unit

The arithmetic logic unit (ALU) performs the CPU's calculations. It handles arithmetic operations (addition, subtraction, multiplication, division) and logical operations (AND, OR, NOT, and comparisons such as equal to or greater than). The ALU is given values from the registers, performs one operation on them, and hands back the result, normally to the accumulator. It does no organising and makes no decisions about what happens next.

The control unit (CU) is the component that does the organising. It fetches each instruction from main memory, decodes it to work out what is required, then sends control signals telling the other components what to do and when. The control unit processes no data itself: it directs. If an instruction says add two numbers, the CU arranges for the right values to reach the ALU and for the ALU to perform the addition.

COMMON EXAM MISTAKE

Writing that the control unit executes instructions or performs calculations. The CU decodes instructions and sends control signals; the ALU performs the operations. An answer that swaps these two functions cannot earn the marks for either component.

The five registers

Registers are tiny storage locations built into the CPU itself. They are far faster than main memory, and each one holds a single value the CPU needs immediately: the instruction it is working on, an address it is about to use, or a result it has just produced. The syllabus names five, and you need the exact function of each.

KEY TERMS: THE FIVE REGISTERS

Program counter (PC): holds the memory address of the next instruction to be fetched. After each fetch it is updated to point to the following instruction, which is how the CPU moves through a program in order.

Instruction register (IR): holds the instruction currently being decoded and executed. Note the difference: the PC holds an address; the IR holds the instruction itself.

Memory address register (MAR): holds the address of the memory location about to be read from or written to. It answers the question: where in memory?

Memory data register (MDR): holds the data just read from memory, or the data about to be written to it. It answers the question: what value? The MAR and MDR always work as a pair.

Accumulator (AC): holds intermediate results while the ALU works. A running total in a loop, for example, sits in the accumulator between additions.

The three buses

THINK FIRST

To use memory, the CPU must tell RAM which location it wants, move the value itself, and signal whether it is reading or writing. Before you read on: how many separate sets of wires do you think that needs, and would they all carry information in both directions?

A bus is a set of parallel wires that carries information between the CPU and other components, mainly memory. Three buses divide the work between them, and the easiest way to remember the division is: the address bus says where, the data bus carries what, and the control bus says when and how.

- **Address bus:** carries memory addresses from the CPU to memory. It is one-directional, because only the CPU issues addresses; memory never sends an address back.
- **Data bus:** carries the data itself between the CPU and memory. It is bi-directional, because data is read from memory and written to it.
- **Control bus:** carries control and timing signals, such as memory read and memory write, between the control unit and other components.

How the components interact

Knowing each component in isolation earns only some of the marks; this learning statement asks for their interactions. Follow one instruction through the system and the connections become clear.

The program counter holds the address of the next instruction. That address is copied to the memory address register and sent to RAM along the address bus. RAM returns the instruction along the data bus into the memory data register, and from there it moves into the instruction register. The control unit decodes it, then sends signals along the control bus and within the CPU to make it happen: if the instruction is a calculation, the ALU receives the values, performs the operation, and places the result in the accumulator. Meanwhile the program counter has already been updated, so the CPU knows where the next instruction is waiting.

This sequence is the fetch, decode and execute cycle, and A 1.1.5 examines it step by step. For this learning statement, what matters is that you can describe which component holds what, and which bus carries what, at each stage.

Single-core, multi-core and co-processors

So far the page has described one processing unit. Real machines vary in how many they have and in whether specialised helpers sit alongside the main CPU.

PROCESSOR	WHAT IT IS	WHAT IT OFFERS
Single-core	A CPU with one processing core, executing one stream of instructions at a time.	Simplicity. Tasks are handled one after another, so heavy workloads queue up.
Multi-core	A CPU with several complete cores on one chip, each able to run its own thread of instructions.	Parallel processing: several tasks genuinely run at the same time. Two cores do not automatically mean double the speed; software must be written to use them.
Co-processor	A specialised processor that assists the main CPU with one kind of task, such as a graphics processing unit (GPU) for graphics or a floating-point unit (FPU) for decimal arithmetic.	Offloading: the specialist handles work it is built for, faster than the general-purpose CPU could, and frees the CPU for everything else.

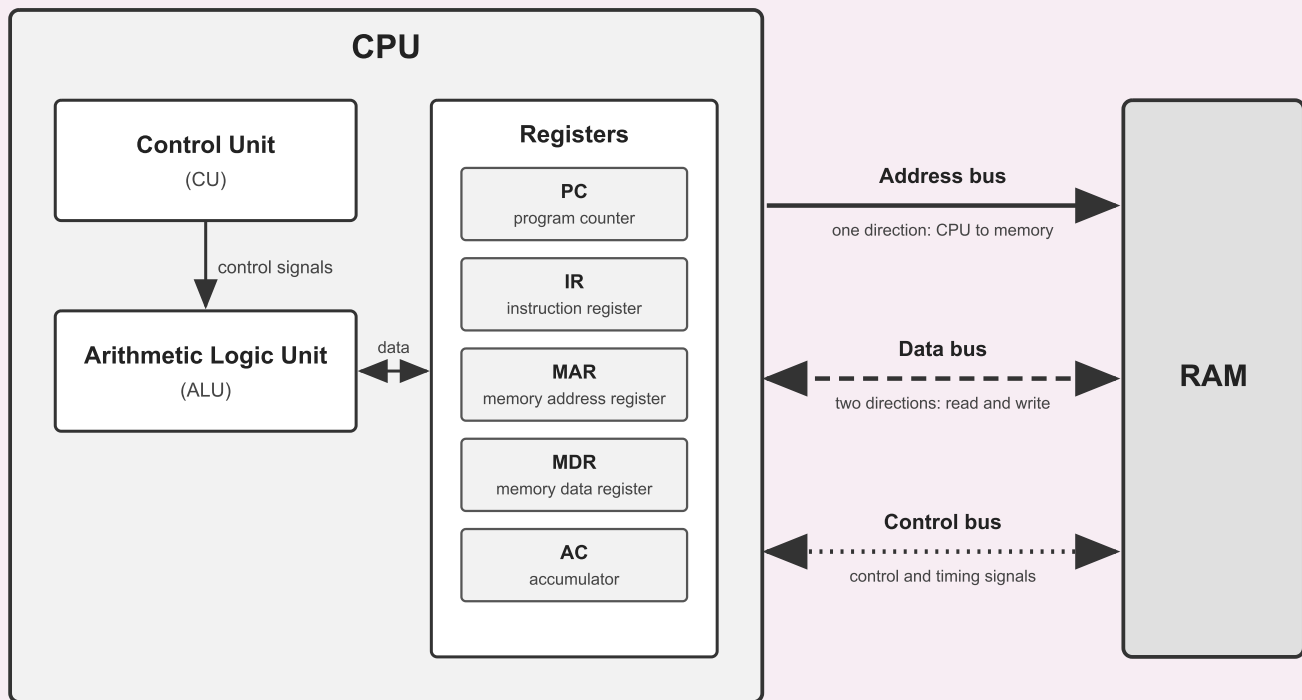
GPUs get a fuller treatment in the next section, A 1.1.2.

The CPU diagram

The syllabus explicitly requires a diagrammatic representation of the relationship between these components, so expect to label, complete or draw a diagram like this one. Practise reproducing it from memory: the units and registers inside the CPU, RAM outside it, and the three buses between them with their directions marked.

EXAM TIP

When drawing the diagram, the directions of the buses are easy marks to gain or lose. The address bus arrow points one way only, from CPU to RAM. The data bus and control bus arrows point both ways. A diagram with three identical double-headed arrows tells the examiner you have memorised a shape, not understood the system.



Main CPU components and the buses connecting the CPU to main memory (RAM)

Figure 1. The main CPU components and the buses connecting the CPU to main memory (RAM).

Summary

Each item below is named in the learning statement, so each is assessable in the exam. Use this table to check you can state the function of every one.

SUMMARY REFERENCE

COMPONENT	FUNCTION
ALU	Performs arithmetic operations (add, subtract, multiply, divide) and logical operations (AND, OR, NOT, comparisons).
Control unit (CU)	Fetches and decodes instructions; sends control signals directing the other components. Processes no data itself.
Program counter (PC)	Holds the address of the next instruction to fetch; updated as the program advances.
Instruction register (IR)	Holds the instruction currently being decoded and executed.
MAR	Holds the memory address about to be accessed (where).
MDR	Holds the data read from or being written to memory (what).
Accumulator (AC)	Holds intermediate results of ALU operations.
Address bus	Carries addresses from CPU to memory. One direction.
Data bus	Carries data between CPU and memory. Both directions.
Control bus	Carries control and timing signals between the CU and other components.
Single-core	One core; one instruction stream at a time.
Multi-core	Several cores on one chip; threads run in parallel.
Co-processor	Specialist (GPU, FPU) that offloads specific work from the main CPU.

Flashcards

Use this Quizlet to learn all the key terms in this section.



Ready to play?

Match all the terms with their definitions as fast as you can. Avoid wrong matches, they add extra time!

Start game

[View this flashcard set](#)

Choose a Study Mode 

A 1.1.1 Quiz

Test yourself on this section to see how well you understand the functions and interactions of the main CPU components.

1

Which CPU component directs the flow of data and instructions inside the processor?

A. ☐ Accumulator (AC)

B. ☐ Control Unit (CU)

C. ☐ Memory Address Register (MAR)

D. ☐ Arithmetic Logic Unit (ALU)

The control unit decodes each instruction and sends control signals telling the other components what to do and when. The ALU is the strongest distractor, but it performs operations on command; it does not direct anything.

2

What is the primary function of the Arithmetic Logic Unit (ALU)?

A. ☒ Performing arithmetic and logical operations

B. ☐ Storing the results of calculations

C. ☐ Fetching instructions from memory

D. ☐ Decoding instructions

The ALU performs arithmetic (add, subtract, multiply, divide) and logical operations (AND, OR, NOT, comparisons). Storing results is the accumulator's job; fetching and decoding belong to the control unit.

3

Which register holds the current instruction being executed by the CPU?

A. ☐ Accumulator (AC)

B. ☒ Instruction Register (IR)

C. ☐ Memory Data Register (MDR)

D. ☐ Program Counter (PC)

The IR holds the instruction itself while it is decoded and executed. The PC is the strongest distractor: it holds the address of the next instruction, not the instruction itself.

4

What is the role of the Program Counter (PC)?

- A. ☐ To store the result of the last calculation
- B. ☐ To hold the address of the data being accessed
- C. ☐ To decode the current instruction
- D. ☒ To store the address of the next instruction to be fetched

The PC always points ahead: it holds the address of the next instruction so the CPU knows where to fetch from.

5

Which register acts as a temporary storage location for data being transferred to and from memory?

- A. ☐ Memory Address Register (MAR)
- B. ☒ Memory Data Register (MDR)
- C. ☐ Instruction Register (IR)
- D. ☐ Accumulator (AC)

The MDR holds the data moving in or out of memory. The MAR is the strongest distractor: it holds the address being accessed, not the data. Where versus what.

6

What is the main function of the address bus?

- A. ☐ To transmit control signals
- B. ☒ To carry memory addresses from the CPU to memory
- C. ☐ To carry data between the CPU and memory
- D. ☐ To connect the CPU to peripherals

The address bus carries addresses, and in one direction only: from CPU to memory. Data travels on the data bus and control signals on the control bus.

7

Which type of bus carries data between the CPU and memory?

- A. ☐ Control bus
- B. ☒ Data bus
- C. ☐ Address bus
- D. ☐ System bus

The data bus carries the values themselves, in both directions, because data is both read from and written to memory.

8

What is a key characteristic of a co-processor?

- A. ☐ It replaces the main CPU.
- B. ☐ It handles all the processing tasks.
- C. ☒ It assists the main CPU with specific tasks.
- D. ☐ It controls the input/output devices

A co-processor is a specialist, such as a GPU for graphics or an FPU for floating-point arithmetic. It offloads particular work from the main CPU; it never replaces it.

9

How does a multi-core processor differ from a single-core processor?

- A. ☐ A multi-core processor consumes less power
- B. ☒ A multi-core processor has multiple processing units, enabling parallel processing.
- C. ☐ A multi-core processor has more cache memory.
- D. ☐ A multi-core processor has a higher clock speed.

The defining difference is multiple cores, each able to run its own thread at the same time. Clock speed, cache size and power consumption all vary between specific processors, but none of them defines the distinction.

10

Which of the following is NOT a common type of co-processor?

- A. ☐ Network Processing Unit (NPU)
- B. ☐ Floating-Point Unit (FPU)
- C. ☒ Central Processing Unit (CPU)
- D. ☐ Graphics Processing Unit (GPU)

GPUs, FPUs and network processing units all assist a main processor with specialised work. The CPU is the main processor itself: by definition it cannot be its own co-processor.

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